Comparative study of jet-substructure quark/gluon discriminants in photoproduction at the EIC and HERA

Siddharth Narayan Singh*
Brandeis University, Waltham, US

R. Aggarwal†
USAR and USBAS, Guru Gobind Singh Indraprastha University, East Delhi Campus, 110092, India

M. Kaur‡
Department of Physics, Panjab University, Chandigarh 160014, India
Department of Physics, Amity University, Punjab, Mohali 140306, India

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Abstract

We compare three jet-substructure quark/gluon discriminators in ep photoproduction — the integrated jet shape $\Psi(r = 0.3)$, the k_T exclusive subjet multiplicity n_{subjets}, and the iterated soft-drop multiplicity n_{SD} — on identical anti-k_T jets from PYTHIA 8 events at $\sqrt{s} = 64$–300 GeV, spanning the EIC design energies and a HERA reference. The integrated jet shape separates gluon from quark jets best at every energy, with ROC areas of 0.75–0.77 against 0.66–0.68 for n_{SD} and 0.64–0.66 for n_{subjets}. The ranking of the two counting observables is not fixed: pooled over pseudorapidity n_{SD} leads, but binned in pseudorapidity n_{subjets} leads in every bin measured. We trace this to the resolution scale of n_{subjets}, which is set by the jet energy rather than its transverse momentum and therefore tightens as $\cosh\eta$. Carrying no tagging parameter beyond the choice of r, and directly comparable to archival HERA data, the integrated jet shape is well matched to first-generation EIC measurements.

I. INTRODUCTION

Telling a quark jet from a gluon jet is one of the oldest problems in jet substructure [1, 2]. At leading logarithmic accuracy the two radiate in proportion to the colour Casimir of the parton that made them, and the ratio $C_A/C_F = 2.25$ sets how much they can differ [3, 4]. Any observable built from the energy and angular distribution inside a jet inherits some share of that difference, which is why so many of them have been proposed. The list runs from the jet shapes and subjet multiplicities measured at LEP and HERA [5–8], through grooming observables such as the iterated soft-drop multiplicity [9, 10] and generalised angularities [11], to deep networks that read the jet constituents directly [12–14]. Which one is best has no single answer: it depends on the collider, the energy range, how much data there is, and how much weight one puts on being able to interpret the result.

The Electron-Ion Collider (EIC), under construction at Brookhaven National Laboratory [15, 16], will be a cleaner place to study jet substructure than a pp collider, and jets are expected to play a major role in its programme [17, 18]. Photoproduction there sits in a particular corner of phase space: jets are softer than at HERA, direct and resolved photon processes mix in proportions that shift smoothly with $\sqrt{s}$, and there is far less underlying-event activity than in pp collisions. That makes it a good setting both for studying jet substructure on its own terms and for testing discriminators before they are used somewhere busier. It also comes with a constraint. The EIC has no data yet, so any choice that needs tuning has to be justified without a calibration to fall back on.

This paper asks the tagging question again, for EIC photoproduction specifically. We compare three substructure observables — the integrated jet shape $\Psi(r)$, the k_T exclusive subjet multiplicity n_{subjets}, and the iterated soft-drop multiplicity n_{SD} — on the same jets, reconstructed the same way, at four centre-of-mass energies covering HERA and the three EIC design points.

There are two reasons to do this. The first is that $\Psi(r)$ is the observable our companion photoproduction study [19] relies on: it is what the thin/thick classifier cuts on to build the quark- and gluon-enriched samples reported there. That choice was made so the results could be compared directly with the published HERA-ZEUS measurements, which is a good reason, but it was never checked against the alternatives at EIC energies. The second is that n_{SD} is widely expected to beat n_{subjets}, because at leading logarithmic accuracy its count approaches the C_A/C_F ratio per hard splitting. Whether that expectation survives in the soft, low-E_T EIC regime, and how either counting observable fares against a continuous one like $\Psi(r)$, is what we set out to measure.

The integrated jet shape wins at every energy we looked at, and the ordering $\Psi(r = 0.3) > n_{\text{SD}} > n_{\text{subjets}}$ holds in every sample. The ordering of the two counting observables, however, turns over once the measurement is binned in pseudorapidity rather than pooled. We trace that to the resolution scale of n_{subjets}, which is normalised to the jet energy rather than its transverse mo-

* siddharthsingh@brandeis.edu
† ritu.aggarwal1@gmail.com
‡ manjit@pu.ac.in

mentum and so tightens as $\cosh\eta$, and we explain the win of $\Psi(r)$ in terms of what the observables do with the jet: one integrates over it, the others count pieces of it. We then say what this means for a first EIC measurement.

The paper is laid out as follows. Section II covers the event generation and jet reconstruction, pointing to the companion paper [19] for the details. Section III defines the three observables. Section IV sets out how the ROC curves are built and what we measure from them. Section V gives the results: per sample, across energies, and as a function of the pseudorapidity binning. Section VI interprets them and draws out the implications for the EIC. Section VII concludes.

II. EVENT GENERATION AND JET RECONSTRUCTION

We generate events with PYTHIA 8.312 [20] using asymmetric ep beams. The photon is quasi-real (`PDF:beamB2gamma = on`): we require $Q^2 < 1$ GeV^2, which is what makes the process photoproduction. PYTHIA mixes the direct and resolved contributions on its own (`Photon:ProcessType = 0`); `HardQCD:all` provides the Born $2 \to 2$ matrix elements for the resolved photon and `PhotonParton:all` those for the pointlike one. We set the multiparton-interaction parameter $p_{T0}^{\mathrm{ref}} = 3.0$ GeV everywhere.

We simulate four beam setups — one HERA reference and the three EIC design energies — with 10^6 events each. Table I lists them.

Table I. Beam configurations and dijet yields inside the analysis window $-1 \le \eta < 2$. All three columns count jets, not events. The mixed quark–gluon sample N_{GQ} is listed for completeness only: it holds one jet of each flavour per event, so its jets carry no class label and it takes no part in the comparison.

$\sqrt{s}$ [GeV]	E_p [GeV]	E_e [GeV]	$\hat{p}_T^{\min}$ [GeV]	N_{QQ}	N_{GG}	N_{GQ}
63.2	100	10.0	5	14 696	737	14 306
104.9	275	10.0	5	19 685	1 908	19 079
140.7	275	18.0	7	114 696	17 519	120 036
300.3	820	27.5	7	96 005	31 990	118 882

We label each event by the flavours of the two outgoing partons of the hard $2 \to 2$ process, which we read straight from the PYTHIA process code. Two outgoing quarks (codes 112, 114, 116, 121–124 for the resolved photon; 271–273 and 281–283 for boson–gluon fusion) give QQ; two outgoing gluons (111, 115) give GG; one of each (113 for the resolved photon; 274 and 284 for QCD Compton) gives GQ. We checked every generated event: the labelling is self-consistent to better than 99.99% in all four samples, with only $\mathcal{O}(10)$ events per sample leaking in from process 284. We use QQ and GG as the two classes to separate. A GQ event contains one jet of each flavour, so its jets carry no class label of their own, and we drop them.

We build jets with FASTJET 3 [21] using anti-k_T [22] at $R = 1.0$. A final-state particle joins the clustering if $p_T > 0.1$ GeV and $|\eta| < 5$. We keep events with exactly two jets in $-4 < \eta < 4$, the leading jet above $E_T = 10$ GeV and the subleading one above 7 GeV. Asking for exactly two jets removes events with extra hard radiation between the jet directions, which in boson–gluon fusion can come from colour exchange between the outgoing partons.

Reconstruction keeps jets over the full $-4 < \eta < 4$, but every result in this paper is quoted inside $-1 \le \eta < 2$. That window is the union of the four pseudorapidity bins used throughout, so the inclusive and the binned numbers are always drawn from the same jets and the per-bin counts sum exactly to the inclusive ones. Section V D shows why the restriction matters: the behaviour of n_{subjets} changes sharply in the forward region, and mixing that region into an "inclusive" number conflates two different effects.

One point matters more than any other here: *all three observables are computed on the same jets, in the same event loop.* Same jet definition, same cuts, same E_T thresholds, at all four energies. Every jet gives a matched triplet $(\Psi(0.3), n_{\mathrm{SD}}, n_{\mathrm{subjets}})$, so any difference in performance between the three comes from the observables themselves and not from the samples or the reconstruction.

III. SUBSTRUCTURE OBSERVABLES

A. Integrated jet shape $\Psi(r)$

The integrated jet shape is the average fraction of a jet's transverse energy that falls inside a cone of radius r around the jet axis,

$$\Psi(r) \;=\; \frac{1}{E_T^{\mathrm{jet}}} \sum_{i \,\in\, \mathrm{jet}} E_{T,i}\, \Theta(r - \Delta R_i)\,, \qquad (1)$$

where $\Delta R_i = \sqrt{(\Delta\eta_i)^2 + (\Delta\phi_i)^2}$ is how far constituent i sits from the jet axis and Θ is the step function. By construction $\Psi(R) = 1$. We read Ψ at $r = 0.3$, the working point of the companion analysis [19] and the value the archival ZEUS measurements are quoted at [7, 8, 23].

Quark jets are more collimated than gluon jets of the same E_T, so they have the larger $\Psi(0.3)$. The other two observables run the other way, larger for gluons, so we build the ROC for Ψ from $-\Psi(0.3)$ and keep one convention throughout.

$\Psi(r)$ has no threshold anywhere in it. Every constituent contributes, weighted by its transverse energy, and nothing is thrown away.

B. Exclusive k_T subjet multiplicity

We recluster the constituents of each anti-k_T jet with the longitudinally-invariant k_T algorithm [24] at the same $R = 1$, using the pair distance $d_{ij} = \min(p_{T,i}^2, p_{T,j}^2)\,\Delta R_{ij}^2/R^2$. We stop once every remaining pair has $d_{ij} > y_{\rm cut}\,Q^2$, and $n_{\rm subjets}$ is how many protojets are left. We take $y_{\rm cut} = 5\times10^{-4}$, the HERA convention, so the result can be compared with the archival data.

The catch is that $y_{\rm cut}Q^2$ is an *absolute* momentum scale, about 0.2 GeV^2 for a 20 GeV jet. Hard perturbative splittings pass it, but so does soft wide-angle radiation. Section V shows what this costs.

C. Iterated soft-drop multiplicity

The iterated soft-drop multiplicity [9] counts how many splittings along the jet's primary branch pass a momentum-fraction cut,

$$n_{\rm SD}(z_{\rm cut},\beta) \;=\; \sum_{i\,\in\,\text{primary}} \Theta\!\left[\, z_i - z_{\rm cut}\left(\frac{\Delta R_i}{R_0}\right)^{\beta}\right], \quad (2)$$

where a splitting into prongs (p_1, p_2) with $p_{T,1} \geq p_{T,2}$ has

$$z_i \;=\; \frac{p_{T,2}}{p_{T,1}+p_{T,2}}, \qquad \Delta R_i \;=\; \Delta R(p_1,p_2). \quad (3)$$

We recluster the constituents with the Cambridge/Aachen algorithm [25, 26] at $R_0 = R$. C/A clusters purely by angle, so undoing it gives an angle-ordered sequence of splittings — the same ordering a coherent QCD shower produces. We then walk the tree from the root, always following the harder prong, and add one to the counter each time Eq. (2) is satisfied. Unlike the soft-drop groomer [10], we never discard the softer prong and we do not stop at the first splitting that passes. That is what makes the observable *iterated*.

We use $z_{\rm cut} = 0.1$ and $\beta = 0$, so the cut is simply $z > 0.1$ regardless of angle — the modified mass-drop condition of Ref. [27], applied over and over. At leading logarithmic accuracy the emission density along the primary branch is proportional to the colour Casimir of the initiating parton, which gives

$$\frac{\langle n_{\rm SD}\rangle_{\rm gluon}}{\langle n_{\rm SD}\rangle_{\rm quark}} \;\longrightarrow\; \frac{C_A}{C_F} \;=\; 2.25 \quad (4)$$

in the asymptotic limit.

The important difference from $n_{\rm subjets}$ is that the cut is on the dimensionless fraction z, not on an absolute momentum. At $\beta = 0$, $n_{\rm SD}$ therefore barely changes if the jet momentum scale changes. This one property drives most of what separates the two counting observables below.

IV. ROC CONSTRUCTION AND FIGURES OF MERIT

For a discriminant X with the convention that gluon jets have the larger value, we take jets from GG events as signal and jets from QQ events as background. A cut at threshold c keeps every jet with $X \geq c$, and we record what fraction of each class survives it:

$$\varepsilon_{\rm GG}(c) = \Pr(X \geq c \mid \mathrm{GG}), \qquad \varepsilon_{\rm QQ}(c) = \Pr(X \geq c \mid \mathrm{QQ}). \quad (5)$$

The two carry opposite meanings despite the symmetric notation. $\varepsilon_{\rm GG}$ is the *gluon-jet efficiency*: the fraction of the jets we want that the cut keeps, so larger is better. $\varepsilon_{\rm QQ}$ is the *quark-jet mistag rate*: the fraction of the jets we want to reject that pass anyway, so smaller is better. No cut makes one favourable without also moving the other, and that trade-off is what the ROC curve displays. The ROC curve is what $\big(\varepsilon_{\rm QQ}(c), \varepsilon_{\rm GG}(c)\big)$ traces out as c sweeps across the range of X. For the two counting observables c runs over integers, so the curve is piecewise linear with one node per cut you can actually make; for $\Psi(0.3)$ it runs over a fine grid. The area under the curve,

$$\mathrm{AUC} \;=\; \int_0^1 \varepsilon_{\rm GG}\,\mathrm{d}\varepsilon_{\rm QQ}, \quad (6)$$

we get by trapezoidal integration over the nodes sorted by increasing $\varepsilon_{\rm QQ}$. An AUC of 0.5 means no discrimination at all and 1 means perfect separation.

Alongside the AUC we quote the mean separation

$$\mathcal{S} \;=\; \frac{\langle X\rangle_{\rm GG} - \langle X\rangle_{\rm QQ}}{\sqrt{\sigma_{\rm QQ}^2 + \sigma_{\rm GG}^2}}, \quad (7)$$

which only sees the shift in the means, while the AUC also picks up differences in the shape of the distributions. Comparing the two is useful: if $\mathcal{S}$ is small but the per-bin AUC is large, the class separation is being washed out by a spread that carries no class information.

It is worth saying plainly what the ROC adds over the fixed-cut analysis of the companion paper [19]. There, a jet is called thin if $\Psi(0.3) > 0.8$ and thick if $\Psi(0.3) < 0.6$, and the purity of the resulting samples is quoted. That is one operating point. The ROC gives the whole curve, so the purity at any other working point follows immediately: for starting populations $N_{\rm QQ}$, $N_{\rm GG}$ and $N_{\rm GQ}$,

$$\mathrm{Purity}_{\rm GG} = \frac{N_{\rm GG}\,\varepsilon_{\rm GG}}{N_{\rm QQ}\varepsilon_{\rm QQ} + N_{\rm GG}\varepsilon_{\rm GG} + N_{\rm GQ}\varepsilon_{\rm GQ}}. \quad (8)$$

Every efficiency and purity quoted here is an ensemble-level number: it is a fraction within a population of classified jets, not a calibrated probability for any individual jet.

V. RESULTS

A. Discriminant ordering

Figure 1 shows the ROC curves for the three observables at each of the four centre-of-mass energies. The ordering

$$\Psi(r = 0.3) \;>\; n_{\rm SD} \;>\; n_{\rm subjets} \tag{9}$$

holds in every sample. The integrated jet shape reaches AUC values between 0.746 and 0.772, the iterated soft-drop multiplicity between 0.656 and 0.684, and the subjet multiplicity between 0.641 and 0.655. Table II gives the numbers.

Table II. Area under the ROC curve for the three discriminants, inclusive over the analysis window $-1 \le \eta < 2$.

$\sqrt{s}$ [GeV]	$\Psi(r = 0.3)$	$n_{\rm SD}$	$n_{\rm subjets}$
64	0.746	0.656	0.655
105	0.767	0.678	0.645
141	0.758	0.668	0.641
300	0.772	0.684	0.651

B. Trend with centre-of-mass energy

Figure 2 shows the same information against $\sqrt{s}$. All three observables are close to flat. Between 64 and 300 GeV the integrated jet shape moves by 0.026 in AUC, the iterated soft-drop multiplicity by 0.028, and the subjet multiplicity by 0.014. None of them depends strongly on the centre-of-mass energy once the pseudorapidity window is held fixed.

This is worth stating explicitly because it is easy to conclude otherwise. Quoted over the full reconstruction acceptance $-4 < \eta < 4$, the subjet multiplicity appears to collapse with energy, from 0.652 at 64 GeV to 0.526 at 300 GeV. That trend is not an energy effect. Higher $\sqrt{s}$ pushes a larger fraction of the sample into the forward region, and Sec. V D shows that the forward region is precisely where $n_{\rm subjets}$ fails. Restricting all four samples to the same $-1 \le \eta < 2$ removes the apparent trend entirely.

So $n_{\rm SD}$ leads $n_{\rm subjets}$ inclusively at every energy above the lowest, by a roughly constant 0.03 in AUC: the gaps are 0.001, 0.034, 0.028 and 0.033 at 64, 105, 141 and 300 GeV. The next two sections show that this ordering does not survive binning, and explain why.

C. The resolution scale of $n_{\rm subjets}$

The exclusive k_T multiplicity is usually described as counting emissions above a fixed momentum scale. In the standard implementation it does not. The $y_{\rm cut}$ condition is normalised to Q^2, the squared sum of constituent *energies*, so a pair is resolved when

$$\min(p_{T,i},\, p_{T,j})\;\Delta R_{ij} \;>\; \sqrt{y_{\rm cut}}\;E_{\rm jet}. \tag{10}$$

Writing the softer prong as a fraction of the jet, $p_T = z\,p_T^{\rm jet}$, and using $E_{\rm jet} = p_T^{\rm jet}\cosh\eta$, the jet transverse momentum cancels from both sides of Eq. (10) and what remains is

$$z\,\Delta R \;>\; \sqrt{y_{\rm cut}}\,\cosh\eta. \tag{11}$$

So $n_{\rm subjets}$ has neither an absolute momentum threshold nor a scale-invariant one. It has an effective momentum-fraction threshold that grows as $\cosh\eta$. With $y_{\rm cut} = 5\times10^{-4}$ and taking $\Delta R \sim R$, the effective cut runs from $z_{\rm eff} \simeq 0.022$ at $\eta = 0$ to 0.053 at $\eta = 1.5$ and 0.137 at $\eta = 2.5$. $n_{\rm SD}$, by contrast, cuts at $z > z_{\rm cut} = 0.1$ everywhere.

Equation (11) makes a sharp prediction. Counting observables accumulate emissions logarithmically, $\langle n\rangle \propto C_R \ln(1/z)$, so $\langle n_{\rm subjets}\rangle$ should fall linearly in $\ln\cosh\eta$ at fixed p_T. Measured on HERA-300 quark jets with $11 < p_T < 16$ GeV, it does: a linear fit gives $\langle n_{\rm subjets}\rangle = 4.35 - 1.53\,\ln\cosh\eta$ with $r^2 = 0.991$, while $\langle n_{\rm SD}\rangle$ stays between 3.42 and 3.48 over the same range. Binned in jet energy rather than η, $\langle n_{\rm subjets}\rangle$ falls from 4.23 at $E = 10$–20 GeV to 1.50 at 70–120 GeV, while $\langle n_{\rm SD}\rangle$ holds at 3.43–3.50. Energy, not transverse momentum, is what the observable responds to.

This also settles which of the two counting observables carries more information. Where $z_{\rm eff} \ll z_{\rm cut}$, that is in the central region, $n_{\rm subjets}$ resolves emissions roughly five times softer than $n_{\rm SD}$ and so samples a larger slice of the Casimir-weighted radiation pattern. It is the better counter there. Its weakness is not the counting but the moving threshold: pooled over a range of η, the same observable is effectively a different measurement in each part of the range.

D. Pseudorapidity binning

Jets are binned individually here, by their own η. The two jets of a dijet event go to whichever bins they each point into, and need not land in the same one; with a mean separation of about 1.2 in η, usually they do not.

If Eq. (11) is the explanation, then binning in η before building the ROC should give $n_{\rm subjets}$ its discrimination back, because inside a narrow bin the effective threshold is nearly constant. It should do almost nothing to the other two, whose thresholds carry no η at all. Figure 3 shows exactly that. The dotted horizontal lines are the inclusive values. For $\Psi(0.3)$ and $n_{\rm SD}$ the per-bin points sit on those lines. For $n_{\rm subjets}$ they sit well above: at HERA-300 it goes from 0.651 pooled to 0.737–0.743 bin by bin.

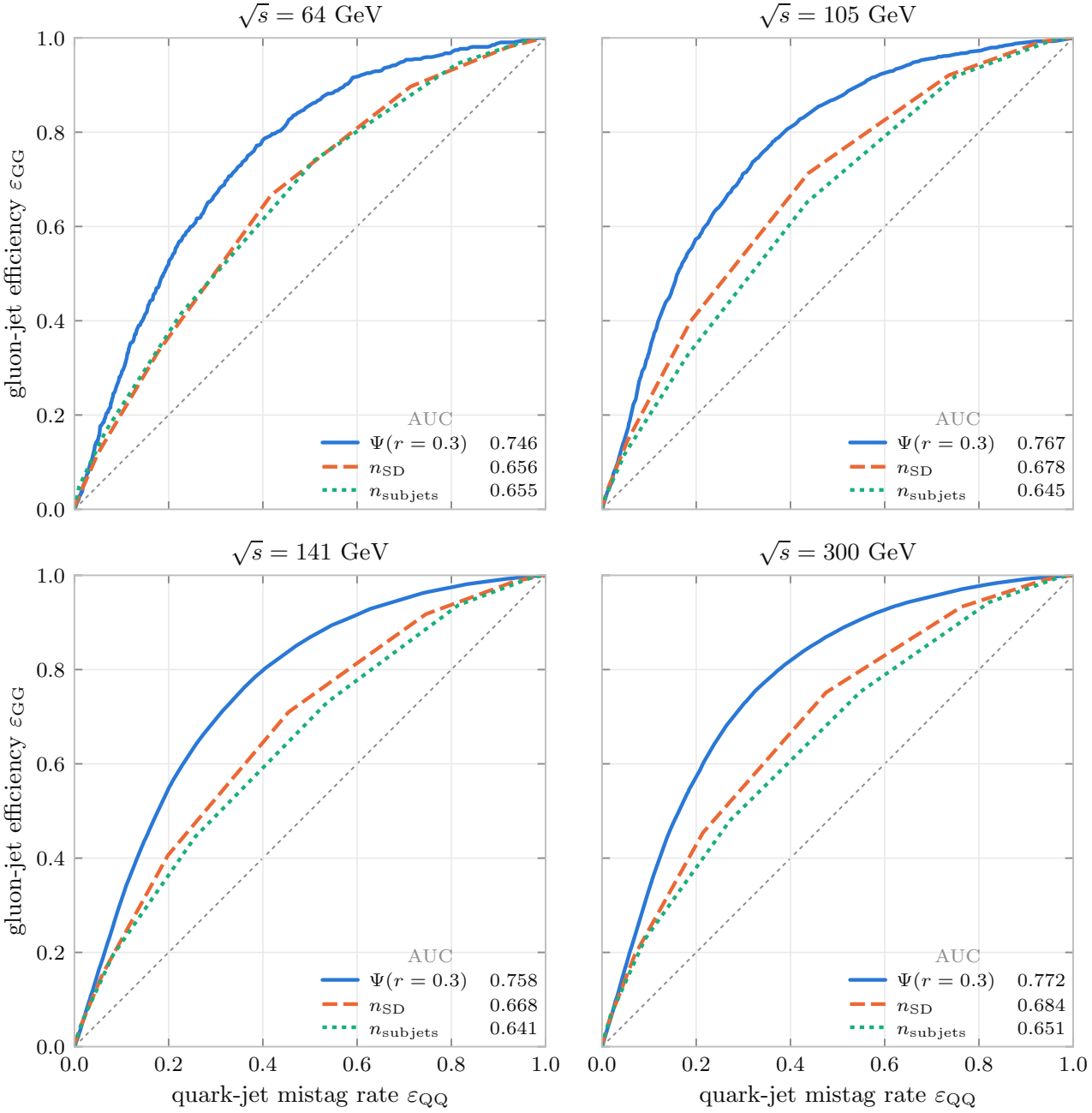


Figure 1. ROC curves for gluon-versus-quark discrimination at the four centre-of-mass energies. Signal is jets from GG events, background is jets from QQ events. The vertical axis is the fraction of gluon jets a cut keeps; the horizontal axis is the fraction of quark jets that pass the same cut. All three observables are evaluated on identical jets (anti-k_T, $R = 1.0$, exclusive two-jet, $E_T > 10$ GeV). The dotted diagonal is the no-discrimination limit. Legend entries give the area under each curve.

This reverses the ordering of the two counting observables. Inclusively n_{SD} is ahead at three of the four energies, by about 0.03. In every one of the fourteen η bins we can measure, $n_{subjets}$ is ahead instead, by between 0.027 and 0.072. There is no exception. Which of the two counting observables looks better is therefore not a property of the observables on their own: it depends entirely on whether the measurement is quoted inclusively or differentially.

The mean separation of Eq. (7) shows the same thing from the other side. Pooled over the window, $n_{subjets}$ at HERA-300 gives $\mathcal{S} = 0.403$ against 0.485 for n_{SD}. Taken bin by bin it gives 0.657, 0.649, 0.671 and 0.667 — larger than n_{SD} everywhere, and remarkably stable, which is what one expects once the effective threshold of Eq. (11) is held nearly fixed inside each bin. The pooled value is depressed not because the classes overlap but because the observable means move between bins: $\langle n_{subjets} \rangle_{QQ}$

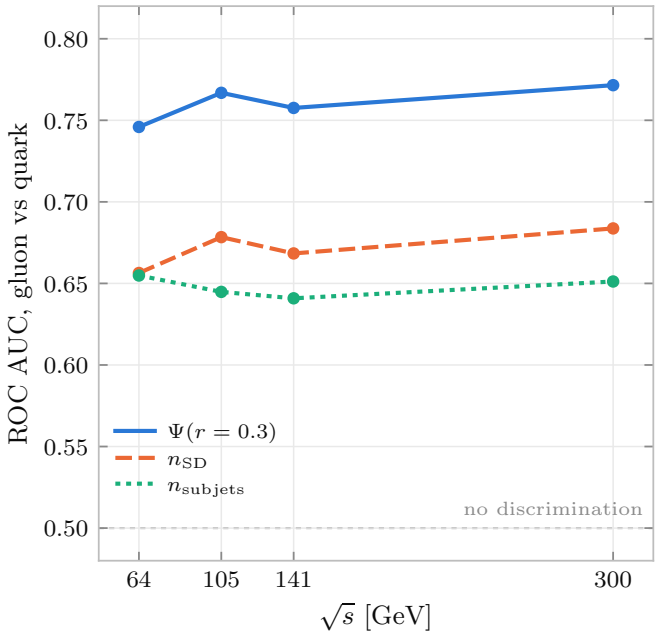


Figure 2. Area under the ROC curve as a function of centre-of-mass energy for the three discriminants, inclusive over $-1 \leq \eta < 2$. With the pseudorapidity window held fixed, all three are close to flat: none varies by more than 0.03 in AUC between 64 and 300 GeV.

runs from 4.25 in the most backward bin to 2.49 in the most forward.

Splitting the pooled variance into within- and between-bin parts,

$$\sigma^2_{\mathrm{pool}} = \underbrace{\langle \sigma^2_{\mathrm{bin}} \rangle_{\mathrm{bins}}}_{\text{within}} + \underbrace{\mathrm{Var}_{\mathrm{bins}}\big(\langle X \rangle_{\mathrm{bin}}\big)}_{\text{between}}, \qquad (12)$$

puts a number on the dilution. For n_{subjets} at HERA-300 the between-bin term is 26.6% of the pooled QQ variance and 38.9% of the pooled GG variance. For n_{SD} it is 0.0% and 0.2%. So between a quarter and two fifths of the width of the pooled n_{subjets} distribution is not class information at all: it is the $\cosh\eta$ of Eq. (11) entering through the back door.

Since that conclusion could be an artefact of one particular choice of bins, we repeated the exercise with uniform bins of varying width over a fixed range $-1 < \eta < 2$, so that the widest case is the genuine pooled limit of the same jets. Table III and Fig. 4 give the jet-weighted mean AUC as the bin width shrinks from 3 (one bin) to 0.25 (twelve bins). $\Psi(0.3)$ and n_{SD} move by no more than 0.007 across that whole range. n_{subjets} climbs by 0.055 at 64 GeV and by 0.094 at 300 GeV, rising monotonically as the bins narrow and flattening off below a width of about 0.5. It is already ahead of n_{SD} at a width of 1.5 and stays ahead as the bins narrow further; only in the fully pooled limit does n_{SD} lead. The effect is therefore not a feature of the binning we happened to pick. It is controlled by how much jet-kinematic spread each bin still contains.

Table III. Jet-weighted mean AUC as a function of η bin width, for uniform bins over $-1 < \eta < 2$. A width of 3 is a single pooled bin. Bins holding fewer than 100 jets of either class are dropped, which at 64 and 105 GeV removes most of the narrowest binning; those two entries rest on limited statistics.

$\sqrt{s}$ [GeV]	width	$\Psi(r = 0.3)$	n_{SD}	n_{subjets}
64	3.00	0.746	0.656	0.655
	1.00	0.748	0.656	0.698
	0.25	0.744	0.666	0.710
105	3.00	0.767	0.678	0.645
	1.00	0.768	0.675	0.702
	0.25	0.767	0.671	0.725
141	3.00	0.758	0.668	0.641
	1.00	0.763	0.665	0.714
	0.25	0.764	0.665	0.731
300	3.00	0.772	0.684	0.651
	1.00	0.776	0.680	0.728
	0.25	0.776	0.679	0.745

$\Psi(0.3)$ is the best of the three either way. It leads in every bin, at every width, at every energy.

E. Colour-factor content

At HERA-300 we measure $\langle n_{\mathrm{SD}} \rangle_{\mathrm{GG}} / \langle n_{\mathrm{SD}} \rangle_{\mathrm{QQ}} = 4.41/3.47 = 1.27$, against the leading-logarithmic expectation $C_A/C_F = 2.25$ of Eq. (4). That is well short of the asymptotic value, and it should be: the LL result assumes plenty of logarithmic room for primary-branch emissions, while a 10–20 GeV jet at $z_{\mathrm{cut}} = 0.1$ only has a handful of splittings that qualify. The counting observable is working far from the regime where its colour-factor argument is sharp, and that puts a ceiling on what any counting tagger can do in this kinematic range.

VI. DISCUSSION

A. Why the integrated observable wins

The three observables differ in how much of the jet they use. Both counting observables boil the jet down to a count of splittings above a threshold. They throw away everything below it and treat everything above it the same. The integrated jet shape sets no threshold at all: every constituent counts, in proportion to its transverse energy, and the radial profile covers hard and soft emissions alike. When there are only a few splittings above threshold to begin with — and Sec. V shows that is the case here — throwing away the rest is expensive. The colour-flow information n_{SD} holds in discrete form is also in $\Psi(r)$, in continuous form, and at these jet energies the continuous form keeps more of it.

Robustness is the second reason. Apart from the choice of r, $\Psi(r)$ has no tagging parameter, and r is less a tuning

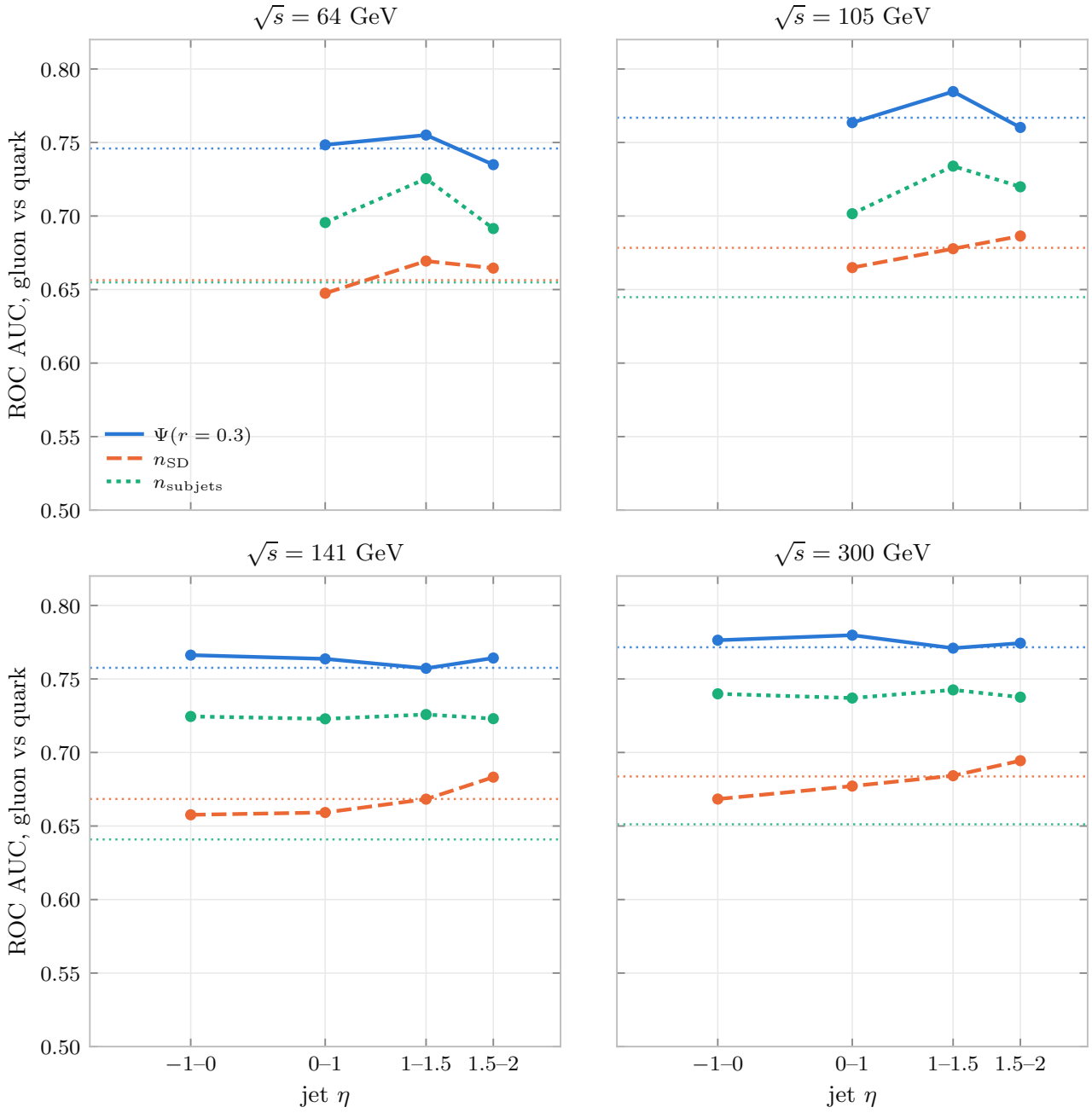


Figure 3. AUC in each pseudorapidity bin (markers) against the inclusive value for the same sample (dotted horizontal lines). $\Psi(0.3)$ and n_{SD} are unaffected by pooling; n_{subjets} loses a large amount of discrimination to it. The $\eta \in [-1, 0)$ bin is absent at 64 and 105 GeV, where the sample has too few jets there.

knob than a choice of where to read a profile you have measured in full. n_{SD} comes with $(z_{\mathrm{cut}}, \beta)$ and n_{subjets} with y_{cut}, and each of those thresholds interacts with the underlying event and the proton remnant in ways that have to be pinned down before the tagger can be trusted. For a first measurement at a facility with no data yet, having fewer such choices to defend is worth a lot.

B. Implications for EIC measurements

These results point to a concrete recommendation. For the first round of EIC jet-substructure measurements, the integrated jet shape is the right choice: it separates the classes best of the three, it holds up across the full EIC energy range and matches the HERA reference, it adds no tagging parameter beyond the choice of r, and it can be compared directly with the archival ZEUS measurements. That last point is shared with n_{subjets}, whose y_{cut}

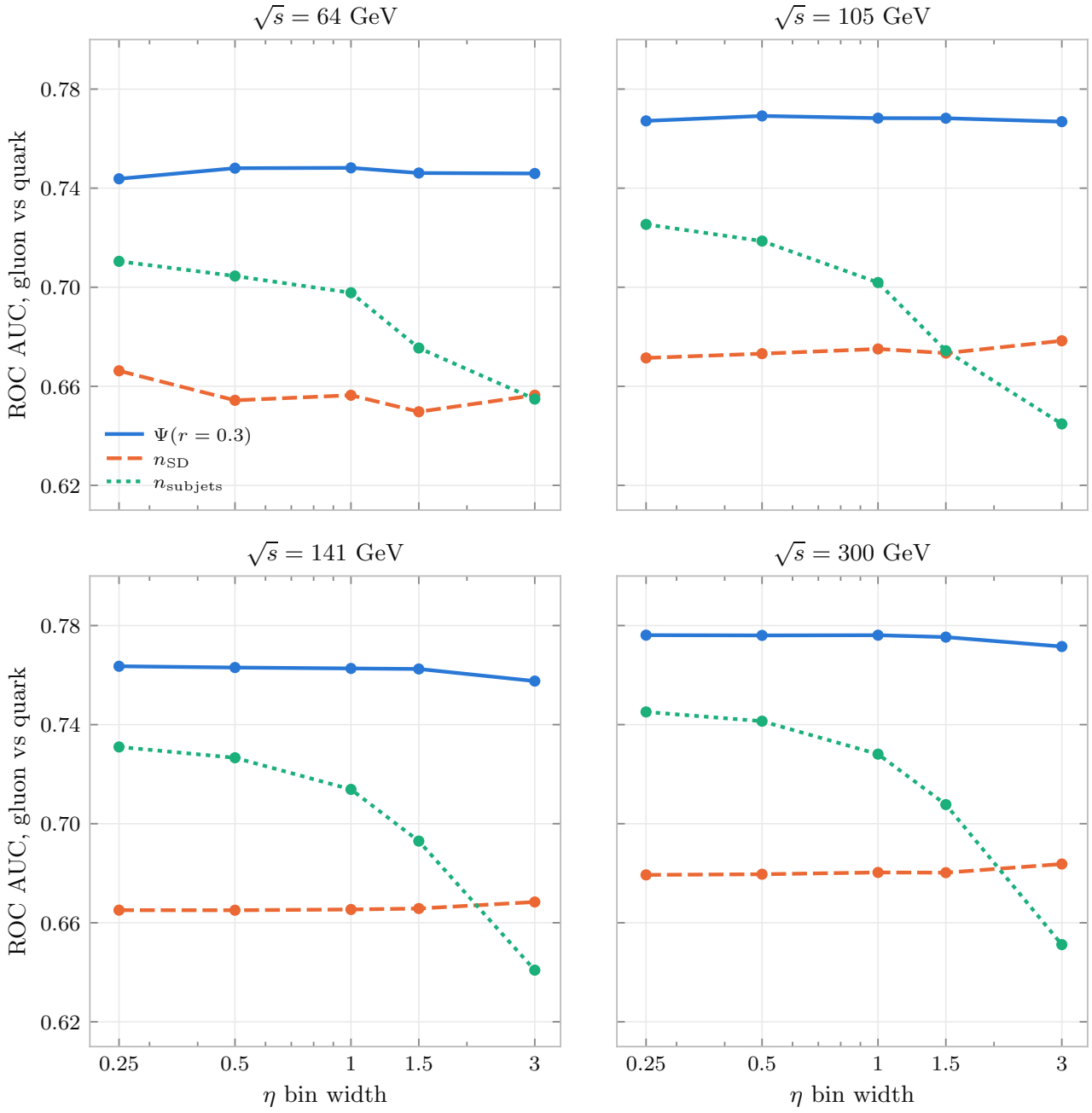


Figure 4. Jet-weighted mean AUC against η bin width, for uniform bins over $-1 < \eta < 2$. A width of 3 is the pooled limit. $\Psi(0.3)$ and n_{SD} are flat; n_{subjets} rises steadily as the bins narrow and overtakes n_{SD} near a width of 1.5.

we fixed to the HERA convention for the same reason, but not with n_{SD}, for which no *ep* measurement exists.

The choice between the two counting observables is not settled in the abstract: it follows from how the measurement is presented. If the result is quoted inclusively in η, use n_{SD}, which is untouched by pooling and leads by about 0.03 in AUC. If it is quoted differentially, which is the more natural way to publish, use n_{subjets}, which leads in every bin we can measure. Section V D shows that bins of width 1.5 or narrower are already enough to put n_{subjets} ahead.

There is also a third option, which this analysis points to but does not pursue. Since the problem is the $\cosh\eta$ in Eq. (11), it can be removed at the source by scaling y_{cut} by $1/\cosh^2\eta$, or equivalently by reclustering in a frame where the jet is central. That would make n_{subjets} behave like a genuinely scale-invariant counter and should recover its per-bin performance without any binning at all. We have not tested it here, but it is a cheap change for any future analysis.

What should not be done is to quote n_{subjets} inclusively over a wide η range without saying so. Its discrimination

power is real, but pooling throws much of it away, and over the full $-4 < \eta < 4$ acceptance the loss is severe.

VII. CONCLUSION

We have compared three quark/gluon discriminants — the integrated jet shape $\Psi(r = 0.3)$, the k_T exclusive subjet multiplicity n_{subjets}, and the iterated soft-drop multiplicity n_{SD} — on the same anti-k_T jets from PYTHIA 8 photoproduction events at HERA and at the three EIC design energies. Four things came out of it.

First, the integrated jet shape separates gluons from quarks best in every sample, with ROC AUCs of 0.746, 0.767, 0.758 and 0.772 at $\sqrt{s} = 64$, 105, 141 and 300 GeV. The iterated soft-drop multiplicity follows at 0.656–0.684, and the subjet multiplicity trails at 0.641–0.655. All three are close to flat in $\sqrt{s}$.

Second, which of the two counting observables is better depends on how the measurement is binned, and the reason is a property of the k_T exclusive algorithm rather than of the physics. Its y_{cut} is normalised to the summed constituent energy, so the jet transverse momentum cancels and the effective momentum-fraction threshold becomes $z_{\text{eff}} = \sqrt{y_{\text{cut}}}\cosh\eta$, rising from about 0.022 centrally to 0.137 at $\eta = 2.5$. n_{SD} cuts at $z > 0.1$ everywhere. The prediction that $\langle n_{\text{subjets}}\rangle$ falls linearly in $\ln\cosh\eta$ is confirmed at $r^2 = 0.991$. Pooled over η, n_{subjets} is therefore a different measurement in each part of the range, and the resulting dilution puts 27–39% of its HERA-300 variance in the between-bin term against under 0.2% for n_{SD}; inclusively n_{SD} wins by about 0.03. Bin in η and the dilution goes away, the ordering reverses, and n_{subjets} leads in all fourteen bins by 0.027 to 0.072 — because where its threshold is soft it resolves emissions about five times softer than n_{SD} does. A scan over bin width shows it already ahead at $\Delta\eta = 1.5$, while $\Psi(0.3)$ and n_{SD} shift by less than 0.007 across the entire scan.

Third, the integrated jet shape wins because it uses the whole radial energy distribution, hard splittings and soft ones alike, instead of counting only what clears a threshold. The colour-flow information n_{SD} carries in discrete form is in $\Psi(r)$ too, in continuous form. And because $\Psi(r)$ has no internal cutoff, it is less exposed to the underlying event and the proton remnant in the low-p_T regime that EIC photoproduction lives in.

Taken together, this is why we would put $\Psi(r = 0.3)$ forward as the headline discriminant for the first round of EIC jet-substructure measurements: it keeps the number of tagging parameters to a minimum and it can be compared directly with the published HERA-ZEUS data. n_{SD} is the natural follow-up, especially once there is real data to calibrate against and enough of it to tune an operating point. We hope what we have done here is useful to the EIC programme in two ways: as a benchmark for comparing taggers, and as the generator-level baseline that detector systematics can later be quoted against.

Appendix A: Supplementary material

1. ROC curves in individual pseudorapidity bins

Section V D quotes per-bin areas but shows the curves only in aggregate. Figures 5 and 6 give the underlying ROC curves bin by bin, at the highest and lowest energies in the study. The reversal discussed in the main text is visible directly: in every panel the n_{subjets} curve lies above the n_{SD} curve, the opposite of what the pooled measurement in Fig. 1 shows.

At $\sqrt{s} = 64$ GeV only three bins are shown. The sample has just ten GG jets with $\eta \in [-1, 0)$, too few for a meaningful ROC, so that bin is dropped; the corresponding QQ count is 690. All four bins are usable at $\sqrt{s} = 300$ GeV.

2. Jet counts per pseudorapidity bin

Table IV lists how many jets enter each bin. Jets are counted individually, not as events: a dijet event contributes its two jets to whichever bins their own pseudorapidities fall in, and the two need not be the same bin. The bins are half-open, $[\eta_{\min}, \eta_{\max})$, so they tile the range $-1 \le \eta < 2$ exactly. The per-bin counts therefore sum to the total over that range with no jet lost and none counted twice, as the closure row shows.

The full-acceptance row is shown only for reference. Reconstruction keeps $-4 < \eta < 4$, but none of the results in this paper use the forward region: every inclusive number is quoted over $-1 \le \eta < 2$, the same window the bins tile. At $\sqrt{s} = 300$ GeV the excluded forward jets are 29% of the QQ sample and 44% of the GG sample, and by Eq. (11) that is exactly where n_{subjets} is worst. Quoting an inclusive number over the full acceptance would therefore mix the binning effect of Sec. V D with a simple change of acceptance, which is why we do not do it.

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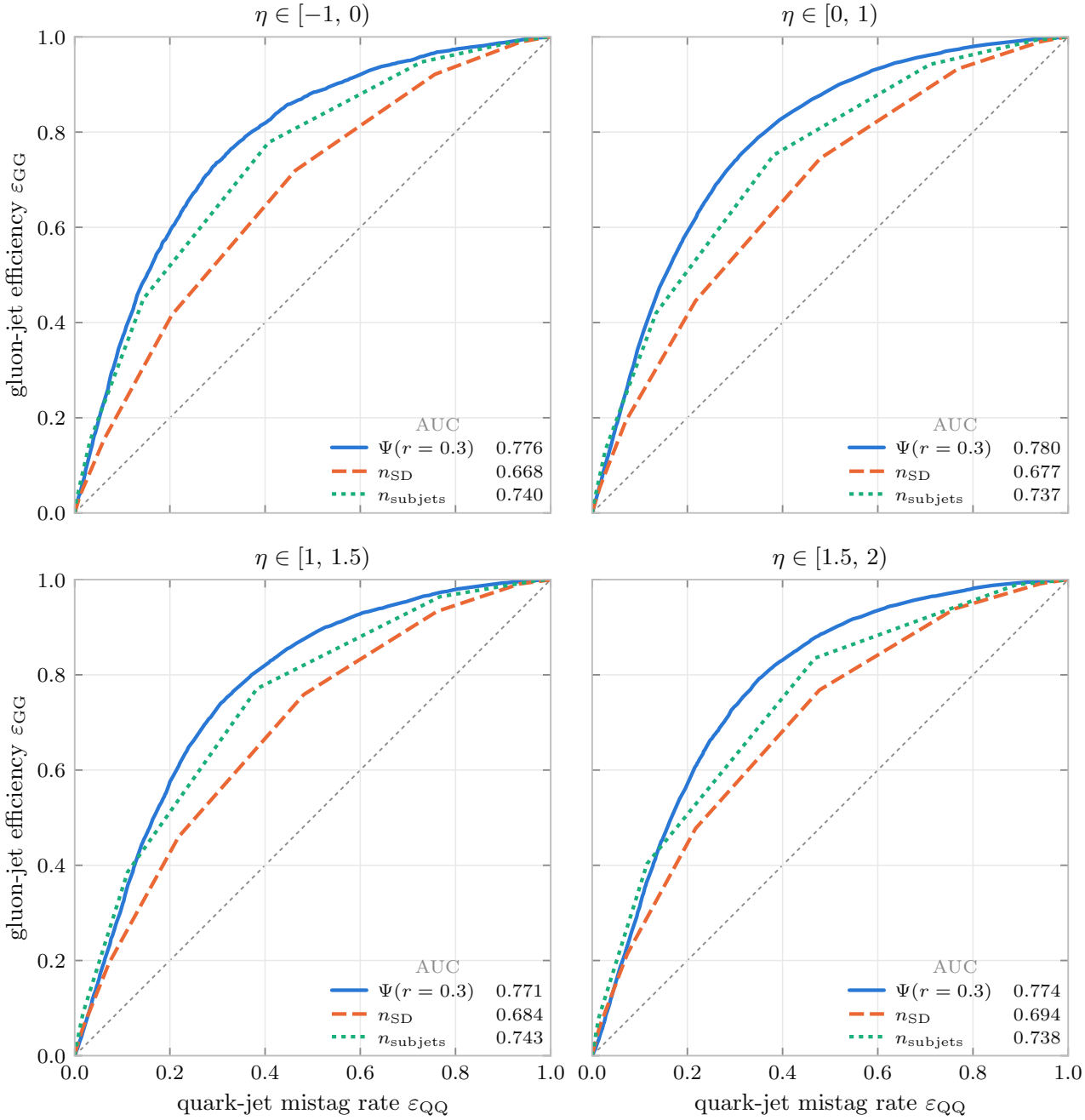


Figure 5. ROC curves in each pseudorapidity bin at $\sqrt{s} = 300$ GeV. Signal is jets from GG events, background is jets from QQ events. Jets are binned individually by their own η. Legend entries give the area under each curve.

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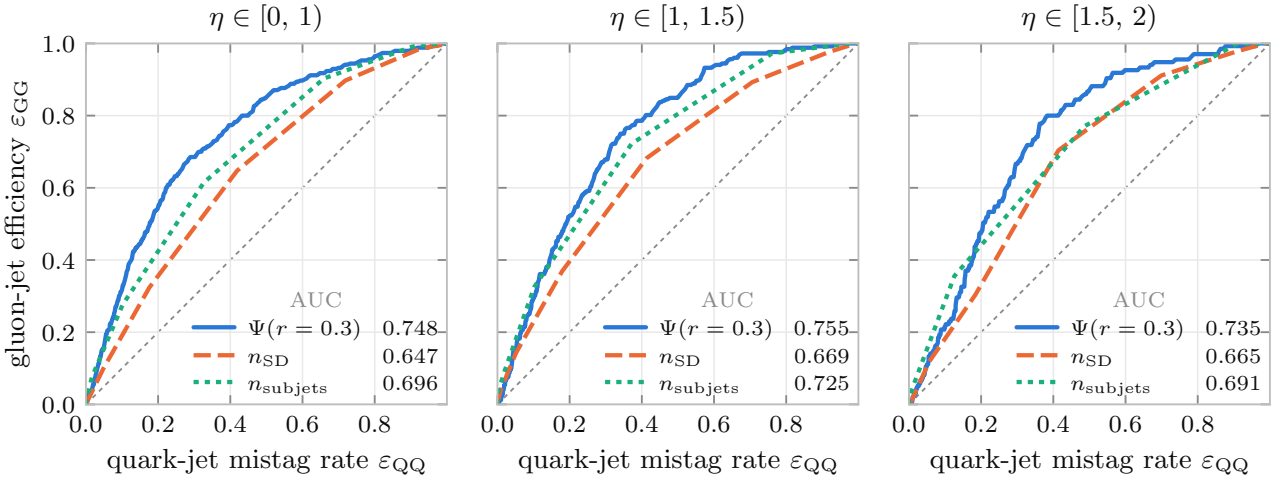


Figure 6. As Fig. 5, at $\sqrt{s} = 64$ GeV. The $\eta \in [-1, 0)$ bin is omitted: it holds only ten GG jets.

Table IV. Jet counts per pseudorapidity bin. "Sum of bins" is the total over the four bins; "total" is a direct count over $-1 \le \eta < 2$. The two agree exactly at every energy. "Full accept." is the reconstruction range $-4 < \eta < 4$, shown for reference only; it is not used anywhere as an inclusive value.

$\sqrt{s}$ [GeV]	bin	N_{QQ}	N_{GG}
64	$[-1, 0)$	690	10
	$[0, 1)$	7 927	340
	$[1, 1.5)$	4 014	252
	$[1.5, 2)$	2 065	135
	sum of bins	14 696	737
	total	14 696	737
	full accept.	15 132	758
105	$[-1, 0)$	617	16
	$[0, 1)$	8 295	601
	$[1, 1.5)$	5 642	612
	$[1.5, 2)$	5 131	679
	sum of bins	19 685	1 908
	total	19 685	1 908
	full accept.	25 258	2 796
141	$[-1, 0)$	20 036	1 538
	$[0, 1)$	50 776	6 634
	$[1, 1.5)$	24 329	4 735
	$[1.5, 2)$	19 555	4 612
	sum of bins	114 696	17 519
	total	114 696	17 519
	full accept.	135 800	22 930
300	$[-1, 0)$	22 897	4 003
	$[0, 1)$	37 961	11 283
	$[1, 1.5)$	18 660	7 965
	$[1.5, 2)$	16 487	8 739
	sum of bins	96 005	31 990
	total	96 005	31 990
	full accept.	134 774	57 592

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